Exercise J-M

1. If $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$, then the value of $\left(a + \frac{1}{a}\right)$ is : **[JEE(Main)jan.2023]**

यदि $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$ है, तो $\left(a + \frac{1}{a}\right)$ का मान है :

(A) $4 - 2\sqrt{3}$ (B) 4 (C) 2 (D) $5 - \frac{3}{2}\sqrt{3}$

Ans. B

Sol. $\Rightarrow \tan 15^\circ + \tan 15^\circ - \tan 15 + \tan 15 = 2a$

$\Rightarrow a = \tan 15^\circ$

Now $\left(a + \frac{1}{a}\right) = \tan 15^\circ + \frac{1}{\tan 15^\circ}$

$= \frac{1 + \tan^2 15^\circ}{\tan 15^\circ} = \frac{2}{\sin 30^\circ} = 4$

2. The value of $36\left(4\cos^2 9^\circ - 1\right)\left(4\cos^2 27^\circ - 1\right)\left(4\cos^2 81^\circ - 1\right)\left(4\cos^2 243^\circ - 1\right)$ is :**[JEE(Main)April,2023]**

$36\left(4\cos^2 9^\circ - 1\right)\left(4\cos^2 27^\circ - 1\right)\left(4\cos^2 81^\circ - 1\right)\left(4\cos^2 243^\circ - 1\right)$ का मान है :**[JEE(Main)April,2023]**

(A) 54 (B) 27 (C) 18 (D) 36

Ans. D

Sol. $4\cos^2\theta - 1 = 4\left(1 - \sin^2\theta\right) - 1 = 3 - 4\sin^2\theta = \frac{\sin 3\theta}{\sin\theta}$

so given expression can be written as

$36 \times \frac{\sin 27^\circ}{\sin 9^\circ} \times \frac{\sin 81^\circ}{\sin 27^\circ} \times \frac{\sin 243^\circ}{\sin 81^\circ} \times \frac{\sin 729^\circ}{\sin 243^\circ}$

$36 \times \frac{\sin 729^\circ}{\sin 9^\circ} = 36$

3. $96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to : **[JEE(Main)April,2023]**

$96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ बराबर है : **[JEE(Main)April,2023]**

(A) 4 (B) 1 (C) 2 (D) 3

Ans. D

Sol. Let $\frac{\pi}{33} = A$

$96\cos A\cos 2A\cos 2^2A\cos 2^3A\cos 2^4A$

$$= 96\frac{\sin 2^5 A}{2^5 \sin A} = \frac{96.\sin\frac{32\pi}{33}}{32\sin\frac{\pi}{33}} = \frac{96.\sin\left(\pi-\frac{\pi}{33}\right)}{32\sin\frac{\pi}{33}} = \frac{96.\sin\frac{\pi}{33}}{32\sin\frac{\pi}{33}} = 3 \text{ Ans.}$$

4. The value of $\tan9° - \tan27° - \tan63° + \tan81°$ is _____. **[JEE(Main)April,2023]**

$\tan9° - \tan27° - \tan63° + \tan81°$ का मान है _____ **[JEE(Main)April,2023]**

Ans. 4

Sol. $\left(\tan 9^\circ + \cot 9^\circ\right) - \left(\tan 27^\circ + \cot 27^\circ\right)$

$$\frac{1}{\sin 9^\circ \cos 9^\circ} - \frac{1}{\sin 27^\circ \cos 27^\circ}$$

$$\frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ}$$

$$\frac{2(4)}{\sqrt{5}-1} - \frac{2(4)}{(\sqrt{5}+1)}$$

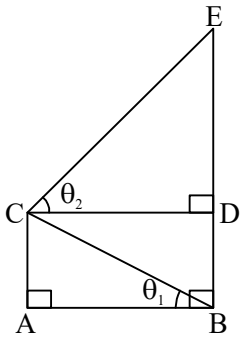
$$\frac{8(\sqrt{5}+1)}{4} - \frac{8(\sqrt{5}-1)}{4}$$

$$2[(\sqrt{5}+1)-(\sqrt{5}-1)]$$

$$= 4$$

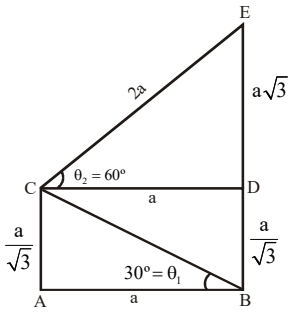
5. In the figure, $\theta_1 + \theta_2 = \frac{\pi}{2}$ and $\sqrt{3}(BE) = 4(AB)$. If the area of ΔCAB is $2\sqrt{3}-3$ unit^2, when $\frac{\theta_2}{\theta_1}$ is the largest, then the perimeter (in unit) of ΔCED is equal to _____. **[JEE(Main)April,2023]**

दी गई आकृति में $\theta_1 + \theta_2 = \frac{\pi}{2}$ तथा $\sqrt{3}(BE) = 4(AB)$ है। यदि ΔCAB का क्षेत्रफल $2\sqrt{3}-3$ वर्ग इकाई है, जब $\frac{\theta_2}{\theta_1}$ अधिकतम है, तो ΔCED का परिमाप (इकाई में) बराबर है : **[JEE(Main)April,2023]**



Ans. 6

Sol.



$$\tan\theta_2 = \frac{DE}{CD} = \frac{BE - AC}{AB}$$

$$\tan\theta_2 = \frac{4}{\sqrt{3}} - \tan\theta_1 \Rightarrow \tan\theta_1 + \tan\theta_2 = \frac{4}{\sqrt{3}} \ldots\ldots$$

$$\text{Now } \theta_1 + \theta_2 = \frac{\pi}{2} \Rightarrow \tan\theta_1 \tan\theta_2 = 1 \ldots (2)$$

Let $\tan\theta_1 = x$ and $\tan\theta_2 = y$ and $\frac{\theta_2}{\theta_1}$ is the largest

$$\Rightarrow x + y = \frac{4}{\sqrt{3}} \qquad x + \frac{1}{x} = \frac{4}{\sqrt{3}} \qquad \sqrt{3}x^2 - 4x + \sqrt{3} = 0$$

$$xy = 1 \qquad \sqrt{3}x\left(x - \sqrt{3}\right) - 1\left(x - \sqrt{3}\right) = 0$$

$$x = \sqrt{3}, x = \frac{1}{\sqrt{3}}$$

$$y = \frac{1}{\sqrt{3}}, y = \sqrt{3}$$

$x = 60^\circ$, $y = 30^\circ$

$$\frac{x}{y} = 2$$

6. For $\alpha, \beta \in (0, \pi/2)$, let $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$ and a real number k be such that $\tan\alpha = k\tan\beta$. Then the value of k is equal to: **[JEE(Main)Jan.2024]**

$\alpha, \beta \in (0, \pi/2)$ के लिए मान $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$ है। माना एक वास्तविक संख्या k के लिए $\tan\alpha = k\tan\beta$ है। तो k का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $-\frac{2}{3}$ (B) 5 (C) –5 (D) $\frac{2}{3}$

Ans. C

Sol. $\frac{\sin(\alpha+\beta)}{\sin\alpha-\beta}=\frac{2}{3}$

$\frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{\sin(\alpha+\beta)-\sin(\alpha-\beta)}=\frac{2+3}{2-3}$

$\frac{2\sin\alpha\cos\beta}{2\cos\alpha\sin\beta}=\frac{5}{-1}$

$\tan\alpha = -5\tan\beta$

7. If $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}$, $\tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan C=\left(x^{-3}+x^{-2}+x^{-1}\right)^{\frac{1}{2}}$, $0<A,B,C<\frac{\pi}{2}$, then A + B is equal to : **[JEE(Main)Jan.2024]**

यदि $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}$, $\tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ तथा $\tan C=\left(x^{-3}+x^{-2}+x^{-1}\right)^{\frac{1}{2}}$, $0<A,B,C<\frac{\pi}{2}$ हैं, तो A + B बराबर है : **[JEE(Main)Jan.2024]**

(A) $\pi - C$ (B) $\frac{\pi}{2}-C$ (C) $2\pi - C$ (D) C

Ans. D

Sol. $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}, \tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$

$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\cdot\tan B}$

$\tan(A+B)=\frac{\frac{1}{\sqrt{x\left(x^2+x+1\right)}}+\frac{\sqrt{x}}{\sqrt{x^2+x+1}}}{1-\frac{1}{\sqrt{x\left(x^2+x+1\right)}}\cdot\frac{\sqrt{x}}{\sqrt{x^2+x+1}}}$

$\tan(A+B)=\frac{1+x}{\sqrt{x}\cdot\sqrt{x^2+x+1}}\times\frac{\left(x^2+x+1\right)}{x^2+x}$

$\tan(A+B)=\frac{\sqrt{x^2+x+1}}{x\sqrt{x}}$ ...(1)

Given $\tan C = \left(\frac{1}{x^3}+\frac{1}{x^2}+\frac{1}{x}\right)^{\frac{1}{2}}$

$\tan C = \frac{\sqrt{1+x+x^2}}{x\sqrt{x}}$ ...(2)

from equation (1) & (2)
$\tan(A+B) = \tan C$
or $A + B = C$

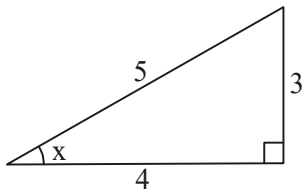
8. If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to : **[JEE(Main)April,2024]**

यदि $\sin x = -\frac{3}{5}$, जहाँ $\pi < x < \frac{3\pi}{2}$, तब $80(\tan^2 x - \cos x)$ बराबर होगा : **[JEE(Main)April,2024]**

(A) 108 (B) 18 (C) 109 (D) 19

Ans. C

Sol.



$\tan x = \frac{3}{4}$, $\cos x = -\frac{4}{5} \Rightarrow 80(\tan^2 x - \cos x) = 80\left(\frac{9}{16}+\frac{4}{5}\right) = 5\times 9 + 16 \times 4 = 109$

9. If the value of $\frac{3\cos 36^\circ + 5\sin 18^\circ}{5\cos 36^\circ - 3\sin 18^\circ}$ is $\frac{a\sqrt{5}-b}{c}$, where a, b, c are natural numbers and gcd(a, c) = 1, then a + b + c is equal to : **[JEE(Main)April,2024]**

यदि $\frac{3\cos 36^\circ + 5\sin 18^\circ}{5\cos 36^\circ - 3\sin 18^\circ} = \frac{a\sqrt{5}-b}{c}$ है, जहाँ a, b, c हैं, तथा gcd(a, c) = 1 है, तो a + b + c बराबर है : **[JEE(Main)April,2024]**

(A) 52 (B) 50 (C) 40 (D) 54

Ans. A

Sol. $$\frac{\frac{3\left(\sqrt{5}+1\right)}{4}+5\left(\frac{\sqrt{5}-1}{4}\right)}{5\left(\frac{\sqrt{5}+1}{4}\right)-3\left(\frac{\sqrt{5}-1}{4}\right)}=\frac{8\sqrt{5}-2}{2\sqrt{5}+8}$$

$$=\frac{4\sqrt{5}-1}{\sqrt{5}+4}\times\frac{\sqrt{5}-4}{\sqrt{5}-4}$$

$$=\frac{20-16\sqrt{5}-\sqrt{5}+4}{-11}$$

$$=\frac{17\sqrt{5}-24}{11} \Rightarrow a = 17, b = 24, c = 11$$

$a + b + c = 52$

10. Let $\left|\cos\theta\cdot\cos\left(60-\theta\right)\cdot\cos\left(60+\theta\right)\right|\le\frac{1}{8},\theta\in\left[0,2\pi\right]$. Then, then sum of all $\theta\in\left[0,2\pi\right]$, where $\cos3\theta$ attains its maximum value of is : **[JEE(Main)April,2024]**

माना $\left|\cos\theta\cdot\cos\left(60-\theta\right)\cdot\cos\left(60+\theta\right)\right|\le\frac{1}{8},\theta\in\left[0,2\pi\right]$ है। तो $\theta\in\left[0,2\pi\right]$ के सभी मानों, जिनके लिए $\cos3\theta$ का मान उच्चतम है, का योग है : **[JEE(Main)April,2024]**

(A) 18π (B) 9π (C) 15π (D) 6π

Ans. D

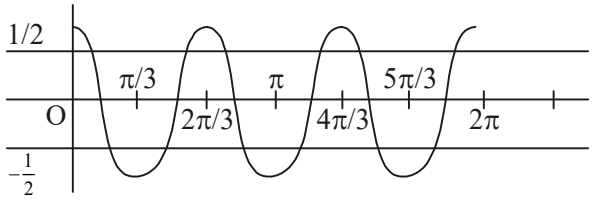
Sol. $$\cos\theta.\cos\left(60-\theta\right)\cos\left(60+\theta\right)=\frac{1}{4}\cos3\theta$$

$$\therefore|\cos3\theta|\le\frac{1}{2}$$

$0 \le 3\theta \le 6\pi$

$$-\frac{1}{2}\le\cos3\theta\le\frac{1}{2}$$

$$\frac{\pi}{6},\frac{\pi}{2},\frac{5\pi}{6},\frac{7\pi}{6},\frac{9\pi}{6},\frac{11\pi}{6}$$



θ satisfying condition are $\frac{\pi}{9},\frac{\pi}{2}+\frac{\pi}{18}+\frac{5\pi}{6}-\frac{\pi}{18},\frac{7\pi}{6}+\frac{\pi}{16},\frac{3\pi}{2}-\frac{\pi}{18},\frac{11\pi}{6}+\frac{\pi}{18}$

$$\sum\theta=\frac{\pi}{9},\frac{\pi}{2}+\frac{\pi}{6}+\frac{7\pi}{6}+\frac{3\pi}{2}+\frac{11\pi}{6}+\frac{\pi}{18}$$

$$=\frac{(2+9+15+21+27+33+1)\pi}{18}=6\pi$$

11. If the range of $f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta},\theta\in R$ is $[\alpha,\beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____. **[JEE(Main)April,2024]**

यदि $f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta},\theta\in R$ का परिसर $[\alpha,\beta]$ है, तब अनंत गुणोत्तर श्रेणी का योग क्या होगा, जिसका प्रथम पद 64 तथा सार्वनुपात $\frac{\alpha}{\beta}$ है, _____ **[JEE(Main)April,2024]**

Ans. 96

Sol. $$f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta}$$

$$f(\theta)=1+\frac{2\cos^2\theta}{\sin^4\theta+\cos^2\theta}$$

$$f(\theta)=\frac{2\cos^2\theta}{\cos^4\theta-\cos^2\theta+1}+1$$

$$f(\theta)=\frac{2}{\cos^2\theta+\sec^2\theta-1}+1$$

$f(\theta)|_{min}=1$

$f(\theta)_{max}=3$

$$S=\frac{64}{1-1/3}=96$$